

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	
Project Title	A Comprehensive Measure of Well-Being (HDI Prediction)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Project Overview	
Objective	The primary objective is to develop a Machine Learning-based HDI Prediction System that accurately classifies countries into Very High, High, Medium, or Low Human Development categories using socio-economic indicators..
Scope	The project includes data collection, preprocessing, machine learning model development, model evaluation, and deployment through a Flask web application that predicts HDI categories based on user inputs.
Problem Statement	
Description	Traditional methods of evaluating Human Development Index require manual analysis of multiple socio-economic indicators, making the process time-consuming and prone to errors. There is a need for an automated system that provides quick and accurate HDI predictions.
Impact	The proposed solution will improve the speed and accuracy of HDI assessment, support evidence-based policymaking, identify development gaps, and help governments and organizations make better decisions for sustainable development.
Proposed Solution	
Approach	Develop a Machine Learning model using Scikit-learn to analyze Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per capita. Deploy the trained model through a Flask-based web application for real-time HDI prediction.
Key Features	Machine Learning-based HDI prediction model. • Real-time prediction of HDI category. • User-friendly web interface using Flask. • Accurate classification into Very High, High, Medium, and Low HDI categories. • Data visualization and prediction reports for better analysis.

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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i3 or above / Equivalent Processor
Memory	RAM specifications	Minimum 4 GB RAM (8 GB Recommended)
Storage	Disk space for data, models, and logs	Minimum 10 GB free disk space
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Development Environment	IDE	Jupyter Notebook, Anaconda Navigator, VS Code or PyCharm.
Data		
Dataset	Source, size, format	Human Development Index (HDI) Dataset UNDP / Kaggle Dataset, CSV format (approximately 200+ country records)